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C204

Final Lab Task 1. Encapsulation

Code:

```
main.py
1 class Car:
2     def __init__(self, color: str, price: float, size: str):
3         self.color = color
4         self.price = price
5         self.size = size.upper()
6     @property
7     def color(self) -> str:
8         return self._color
9
10    @property
11    def price(self) -> float:
12        return self._price
13
14    @property
15    def size(self) -> str:
16        return self._size
17
18    @color.setter
19    def color(self, value: str) -> None:
20        self._color = value
21
22    @price.setter
23    def price(self, value: float) -> None:
24        self._price = value
25
26    @size.setter
27    def size(self, value: str) -> None:
28        if value.upper() not in ['S', 'M', 'L']:
29            raise ValueError("Size must be one of 'S', 'M', 'L'.")
30        self._size = value.upper()
31
32    def __str__(self) -> str:
33        size_descriptor = {'S': 'small', 'M': 'medium', 'L': 'large'}
34        return f"Car ({self._color}) - {self._price:2f} - {size_descriptor[self._size]}"
35
36
37 #Sample 1
38 car1 = Car("red", 20000.00, "M")
39 print(f"Action: Invoking the Car class constructor using Car(\"red\", 20000.00, \"M\").")
40 print(f"Output:\n{car1}")
41
42 #Sample 2
43 car2 = Car("blue", 50000.00, "L")
44 print(f"Action: Invoking the Car class constructor using Car(\"blue\", 50000.00, \"L\").")
45 print(f"Output:\n{car2}")
46
47 #Sample 3
48 car3 = Car("green", 75000.00, "S")
49 print(f"Action: Invoking the Car class constructor using Car(\"green\", 75000.00, \"S\").")
50 print(f"Output:\n{car3}")
51
```

Output:

```

Action: Invoking the Car class constructor using Car("red", 20000.00, "M").
Output:
Car (red) - 20000.00 - medium

Action: Invoking the Car class constructor using Car("blue", 50000.00, "L").
Output:
Car (blue) - 50000.00 - large

Action: Invoking the Car class constructor using Car("green", 75000.00, "S").
Output:
Car (green) - 75000.00 - small

...Program finished with exit code 0
Press ENTER to exit console.
```